

Service Publication

Diesel Engine
16V 396 TB 84
Description and
Operation Manual
M010990/00E

DETROIT DIESEL



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Notes for the user

This Manual consists of 10 Sections, marked by tab-index sheets.

These Sections are listed on this page.

Each Section has its own detailed Table of Contents which lists the Groups and Subgroups.

The Groups in Sections B and G correspond to the Groups in the Spare Parts Catalog and other Product Support publications.

Using the Table of Contents to trace an entry

- Table of Contents (general)
- Table of Contents (section)
- Group / Subgroup
- Heading

Using the Alphabetical Index to trace an entry

- Term in Alphabetical Index
- Group / Subgroup
- Heading

Publication Amendment Service

Please complete and return the Confirmation of Receipt card; you will then receive amendments to your manual.

A	Product Summary Engine summary, technical data
B	Description System and subassembly descriptions
C	Operating Instructions Task descriptions
D	Maintenance Schedule Schedule and maintenance frequency chart
E	Troubleshooting Fault conditions, probable causes, remedies
F	Drawings Engine cross-sections
G	Task Description Inspection and maintenance
K	Preservation Engine preservation
Z	Appendix Supplementary documentation
?	Index Alphabetical index of key terms

Amendment Service!

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M010990/00E

**Description and
Operation Manual**

16V 396 TB84



Postcard

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GERMANY

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Contents

A.000 Product summary - Group A

A.010 Engine summary

Engine model designation

Engine power

Engine layout - cutaway drawings

A.020 Engine side and cylinder designations

Designations and abbreviations

A.030 General specifications

A.040 Main engine dimensions

Engine weight, coolant capacity

Engine inclinations/oil capacities

A.050 Operational data

Pressures

Temperatures

Consumptions

Sequential turbocharging control, limit and setting values

Charge air preheating control, input values



Diesel Engine

Part A Product Summary

The product summary gives details of the engine and contains technical data such as engine power, general engine specifications, engine weight, oil and coolant capacities, settings and operational data.



ENGINE LAYOUT

Key to Engine Model Designation 16 V 396 TB 84

16	= number of cylinders
V	= Vee configuration
396	= engine series - 100 times displacement of one cylinder in liters
T	= exhaust turbocharging
B	= external charge air cooling
8	= engine for marine propulsion
4	= design digit

Engine Power

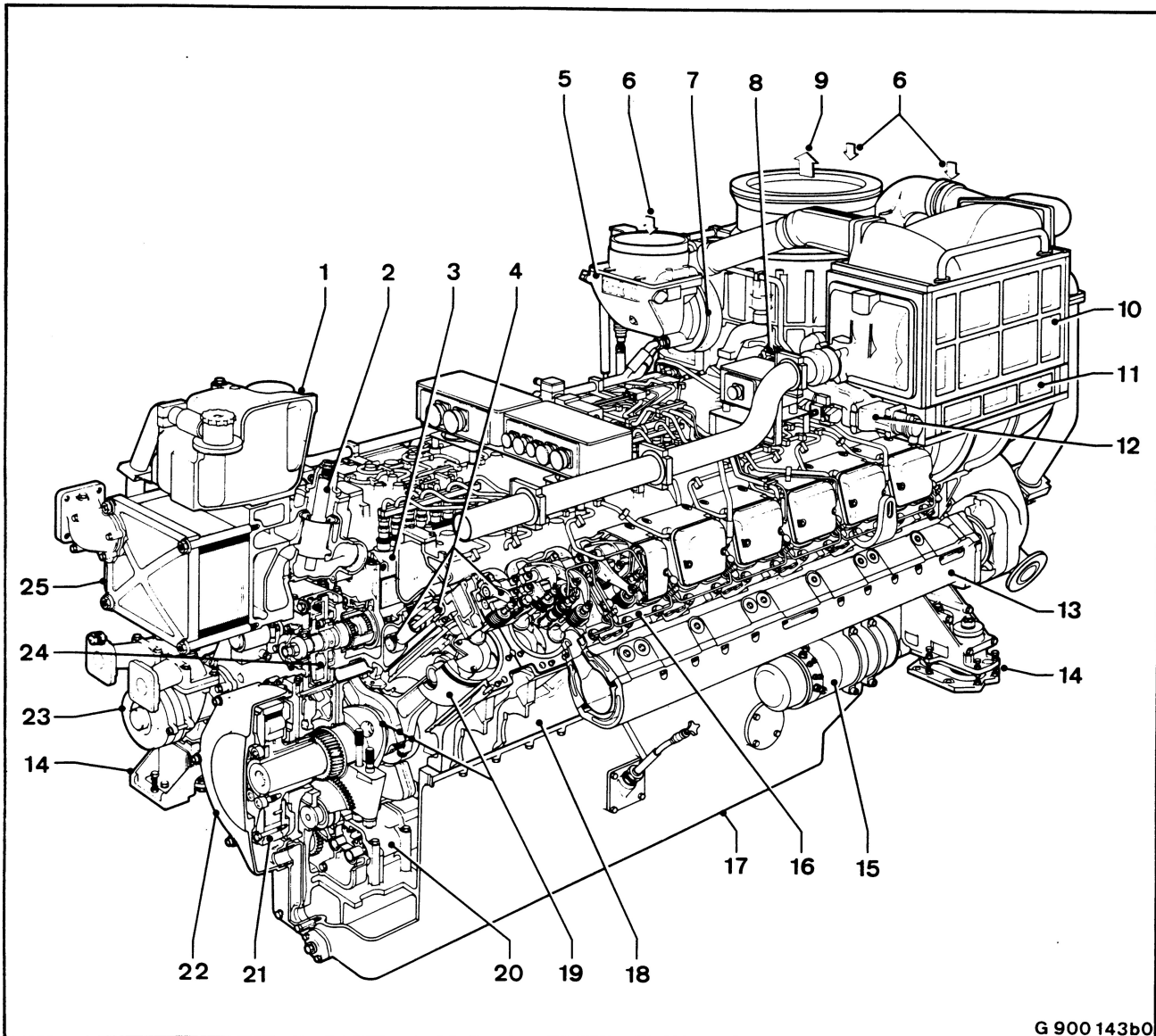
Continuous power
(fuel stop power) 2240 kW at 2000 rpm

Reference conditions

- intake air temperature 25°C
- raw water temperature 25°C
- barometric pressure 1000 mbar

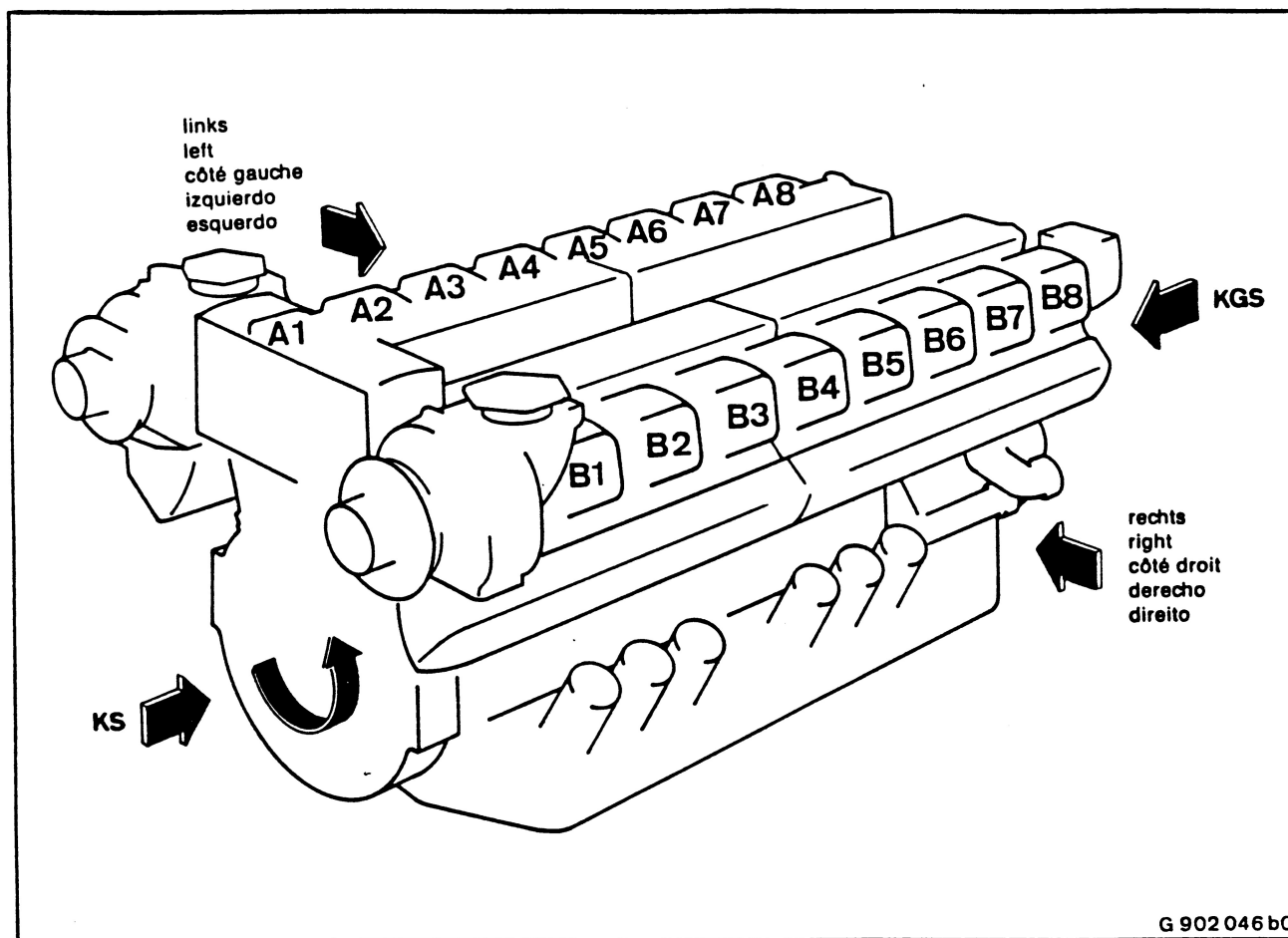


Engine Layout Cutaway Drawing 16 V 396 TB 84/94 (also applies to 12 V)



G 900 143b0

- | | |
|-------------------------------------|-----------------------------|
| 1 Coolant expansion tank | 14 Engine mount |
| 2 Coolant thermostat | 15 Starter |
| 3 Fuel injection pump | 16 Cylinder head |
| 4 Valve gear | 17 Crankcase, lower section |
| 5 Air flow control flap | 18 Crankcase, upper section |
| 6 Air inlet | 19 Running gear |
| 7 Exhaust turbocharger | 20 Engine oil pump |
| 8 Emergency air shut-off flap | 21 Vibration damper |
| 9 Exhaust outlet | 22 Cover |
| 10 Intercooler | 23 Bilge pump |
| 11 Charge air preheater | 24 Fuel injection timer |
| 12 Control for charge air preheater | 25 Coolant cooler |
| 13 Exhaust manifold | |

ENGINE SIDE AND CYLINDER DESIGNATIONS

Engine sides are always designated as viewed from the flywheel end.

The left bank of cylinders is marked "A" and the right bank "B" (according to DIN standard 6265).

The cylinders of each bank are numbered consecutively, starting with No. 1 at the flywheel end.

Numbering of other engine components is also from the flywheel end, starting with No. 1.

Designations and abbreviations

Flywheel end = KS (formerly HKS)

Timing end = KGS (formerly GKS)

Left side (formerly A side)

Right side (formerly B side)

GENERAL SPECIFICATIONS

Operating method	four-stroke cycle, single-acting
Combustion method	direct injection
Mode of supercharging	single-stage sequential exhaust turbocharging
Cooling method	liquid cooling
Cylinder arrangement	90° Vee
Bore	165 mm
Stroke	185 mm
Displacement, cylinder	3.96 liters
Number of cylinders	16
Displacement, total	63.3 liters
Compression ratio	11.0 : 1
Direction of rotation (viewed from flywheel end)	c.c.w. (non-reversible)
Firing order	A1 A3-A5 A2-A8 A7 A4-A6 B5 B2-B8 B3 B4-B6 B1-B7
Fuel injection pressure:	
Setting pressure with new spring	280 to 288 bar
Specified pressure with run-in spring	270 to 278 bar
Minimum pressure with run-in spring	260 bar
Final compression pressure at 120 rpm, 1000 bar and engine coolant at 5°C	19 to 23 bar
Break-away torque at engine coolant temperature of 40°C, allowance for acceleration included	approx. 800 Nm
Cranking torque at engine coolant temperature of 40°C	approx. 750 Nm
Firing speed at engine coolant temperature of 40°C	approx. 120 rpm
Mean piston speed - at 2000 rpm	12.33 m/s
Engine noise	104 dB(A)
Mean free-field level, at a distance of 1 m	
Exhaust noise, undamped	110.5 dB(A)
Free-field level, at a distance of 1 m	